

Playroom Action Analysis

AI-Based Behavioral Analysis for Autism Research

INTRODUCTION

Playroom Action Analysis is a computer vision system for automated analysis of recorded child play sessions. Using YOLO detection and camera-to-room mapping, it visualizes the child's walking path and records time spent in predefined regions of interest.

GOAL

Develop an automated Computer Vision system that analyzes recorded playroom sessions, tracks child movement, detects interactions with predefined regions, and generates structured behavioral reports to support Autism Spectrum Disorder research.

SYSTEM WORKFLOW

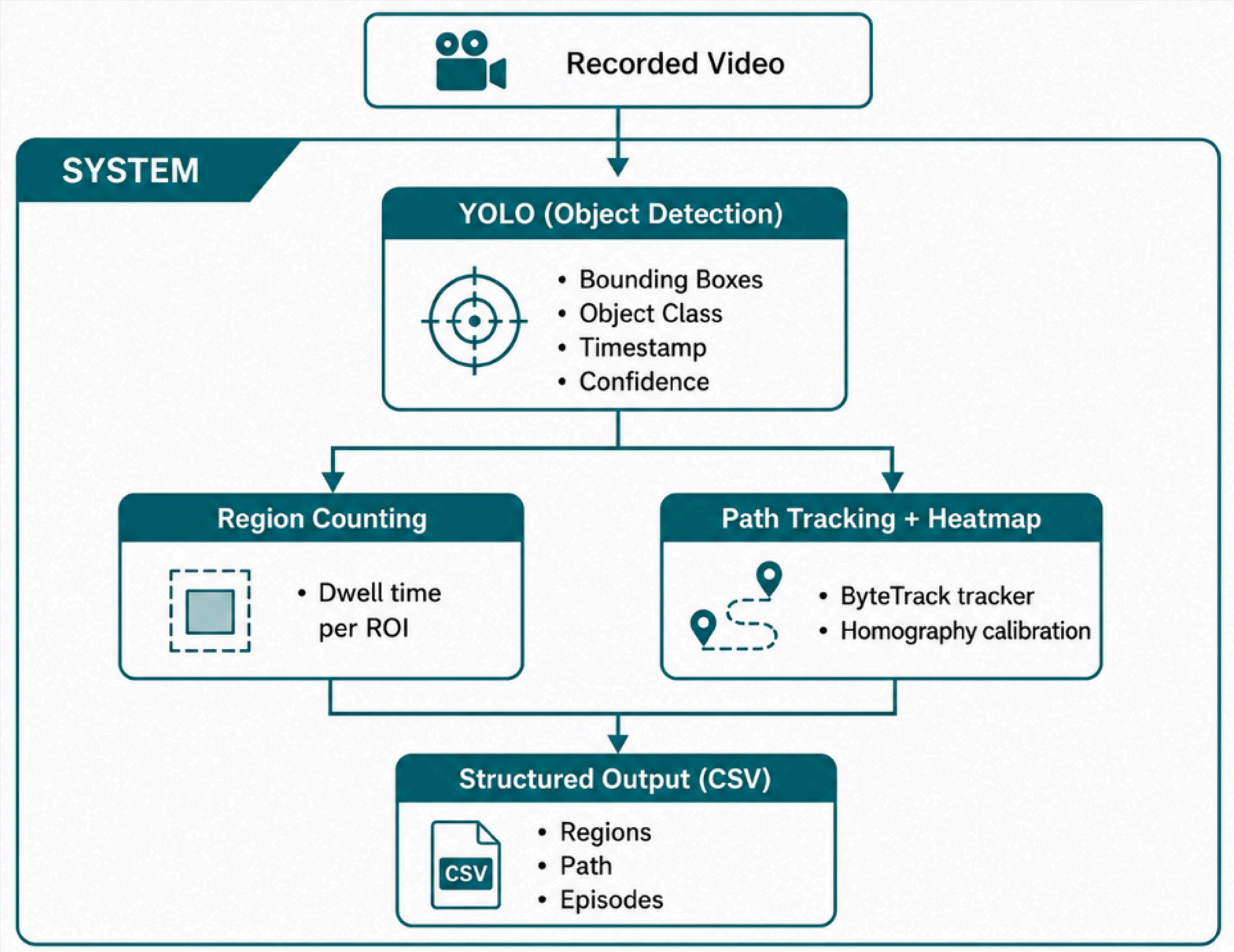


Figure 1: System architecture diagram showing the processing pipeline from recorded video input through YOLO object detection to region counting and path tracking outputs.

SYSTEM GUI

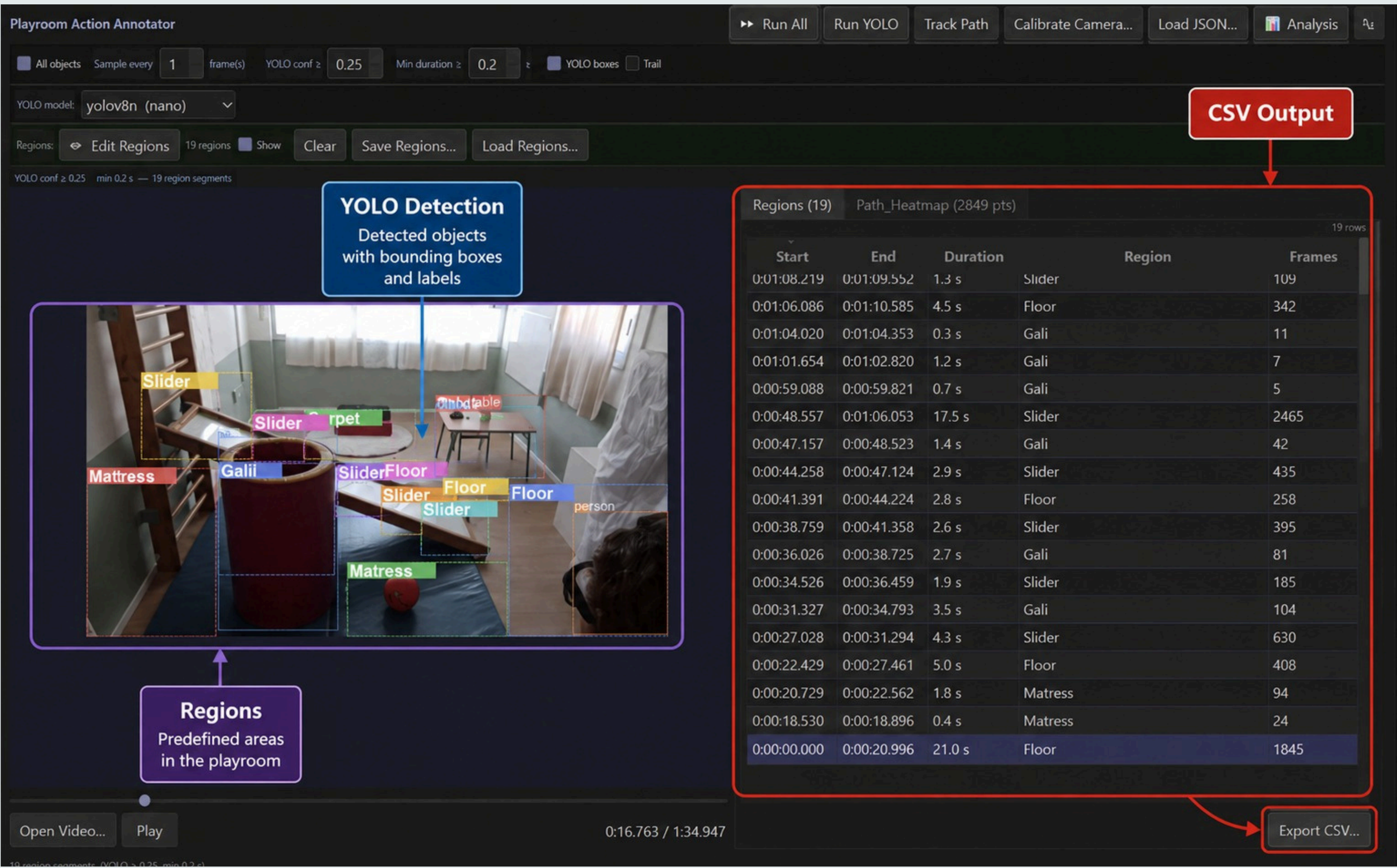
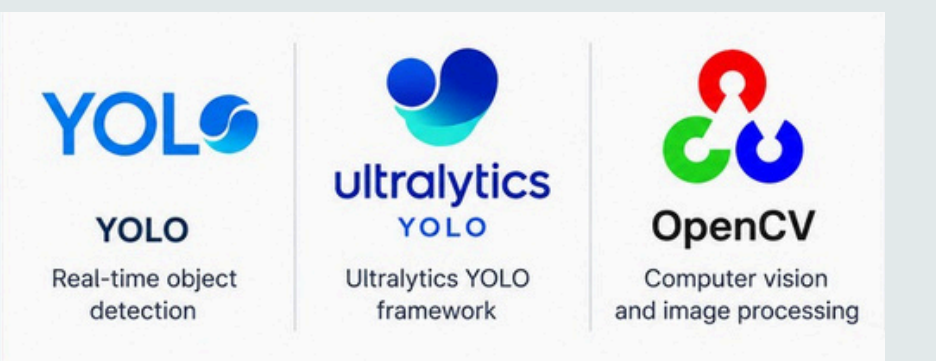


Figure 2: Screenshot of the Playroom Action Annotator interface displaying YOLO detection with bounding boxes and labeled regions overlaid on video playback. to the right is the Region presence table showing detected time segments with start time, end time, duration, and assigned region for each entry.



PATH TRACKING



Figure 3: Top-down path map visualization with the child's tracked walking trajectory and labeled region zones within the playroom.

INFORMATION

Total Distance	41.14 m	Avg Speed	0.433 m/s	Peak Speed	8.131 m/s
Session Duration	1m 35s	Floor Coverage	7.5 %	Active / Stationary	63.2 % / 36.8 %
Active Episodes	5	Stationary Episodes	6	Avg Active Duration	12.0 s
Avg Stationary Dur.	5.8 s	Track Points	2849	Interpolated Points	4

Figure 4: Summary of path tracking metrics including total distance, session duration, average and peak speed, floor coverage, and episode counts.

RESULT

- Reliable child detection using configurable YOLO models.
- Positive feedback from the ASD researchers for pathtracking
- Accurate path tracking and heatmap generation.
- Synchronized playback with automatic region event updates.
- Structured CSV export for behavioral analysis.